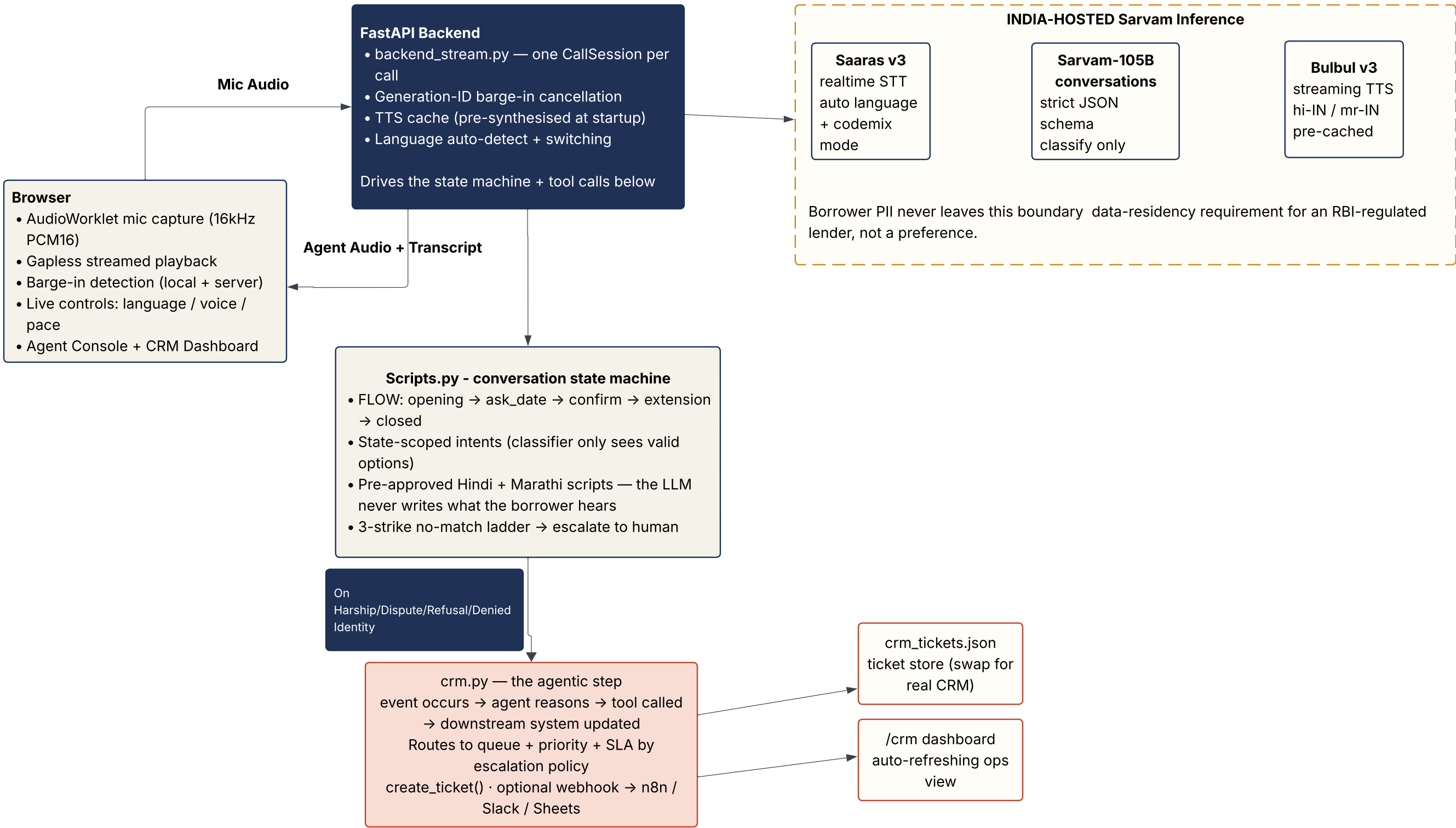


Sovereign Voice Collections Agent - Architecture

- One WebSocket per call
- Streaming STT & TTS
- Scripted responses
- Agentic CRM escalation



One turn, end to end

1. Borrower speaks → streamed to Saaras (server VAD detects end of speech, auto language detect + codemix)
2. Transcript → Sarvam-105B classifies intent against a strict schema (never writes the spoken reply)
3. State machine picks the pre-approved script line (Hindi or Marathi) → Bulbul streams it back, played from cache when known in advance
4. If the turn needs a human, crm.py creates a ticket with queue / priority / SLA / next action → visible on /crm within ~3 seconds

Throughout: mic stays live — a generation-ID system lets the borrower interrupt the agent at any point, discarding stale audio server- and client-side

Production path: swap the browser mic for a SIP trunk (Twilio/Exotel, 8kHz μ law); swap crm.py's JSON store for the lender's real CRM/LOS; move orchestration to Pipecat or LiveKit with Sarvam's official plugins, which already solve reconnection and turn-taking with tuned defaults.